Assignment No: 4

Title: Implement and test a distributed storage system using HDFS and Apache HBase for large- scale data storage.

Steps:

1. **Stop HBase:**

   stop-hbase.sh

2. **Clear old data (Optional):**

   hdfs dfs -rm -r /bank_data

   hdfs dfs -rm -r /hbase

3. **Restart Services:**

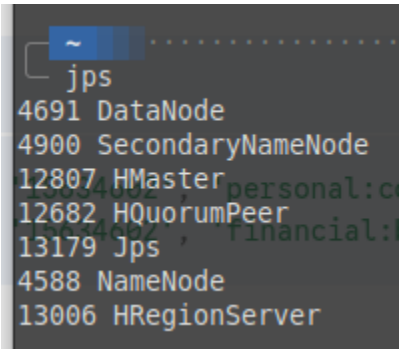
   start-dfs.sh

   start-hbase.sh

Extra step ->

Open the new terminal, write *jps,* hit enter.

Following all processes must be running.



---

4. **Prepare CSV:** Ensure Bank_Customer_Churn_Dataset.csv is in your local folder. (put **Bank_Customer_Churn_Dataset.csv** in **home** directory)

5. **Create HDFS Folder:**
   hdfs dfs -mkdir /bank_data
6. **Upload to HDFS:**
   hdfs dfs -put Bank_Customer_Churn_Dataset.csv /bank_data

Enter the HBase Shell:

hbase shell

7.Create Table:

create 'bank_customers', 'personal', 'financial'

8.Verify Schema:

describe 'bank_customers'

CRUD Operations (The Data)

Inside the HBase Shell:

Insert (Create):

put 'bank_customers', '15634602', 'personal:country', 'France'

put 'bank_customers', '15634602', 'financial:balance', '0.00'

Read (Get):

get 'bank_customers', '15634602'

Update:

put 'bank_customers', '15634602', 'financial:balance', '500.00'

Delete a Cell:

delete 'bank_customers', '15634602', 'financial:balance'

Scan Everything:

scan 'bank_customers'

Deleting the Table

Inside the HBase Shell:

Disable First:

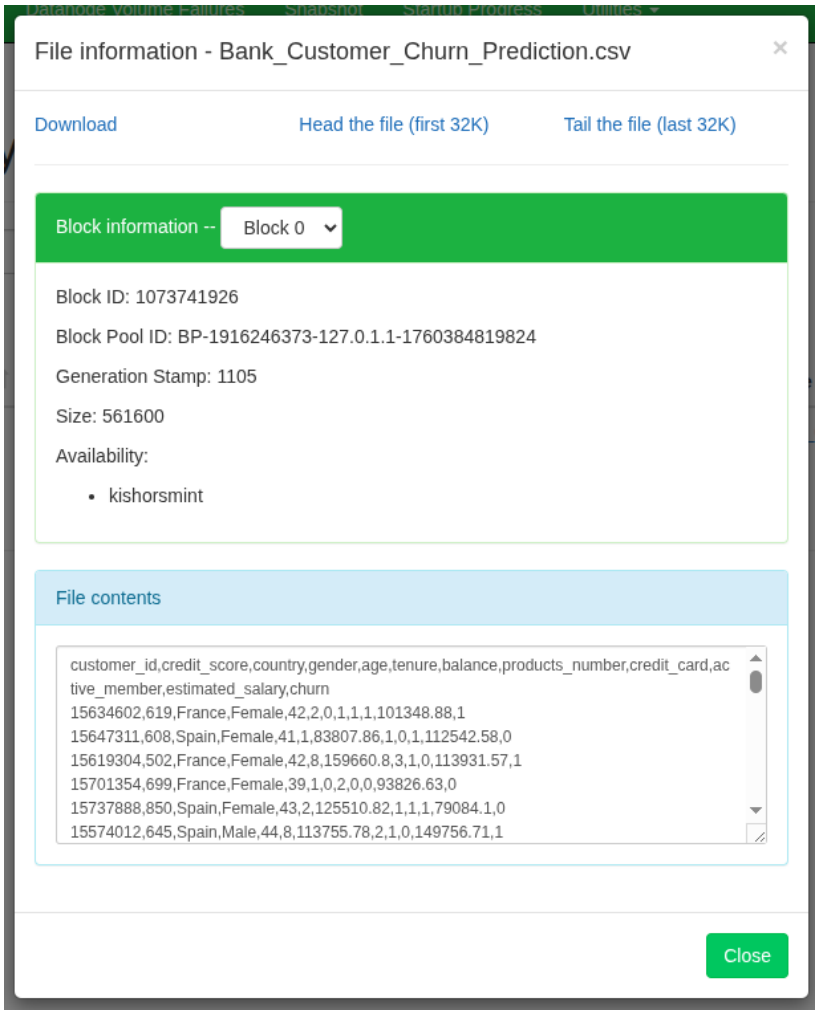
disable 'bank_customers'

Drop:

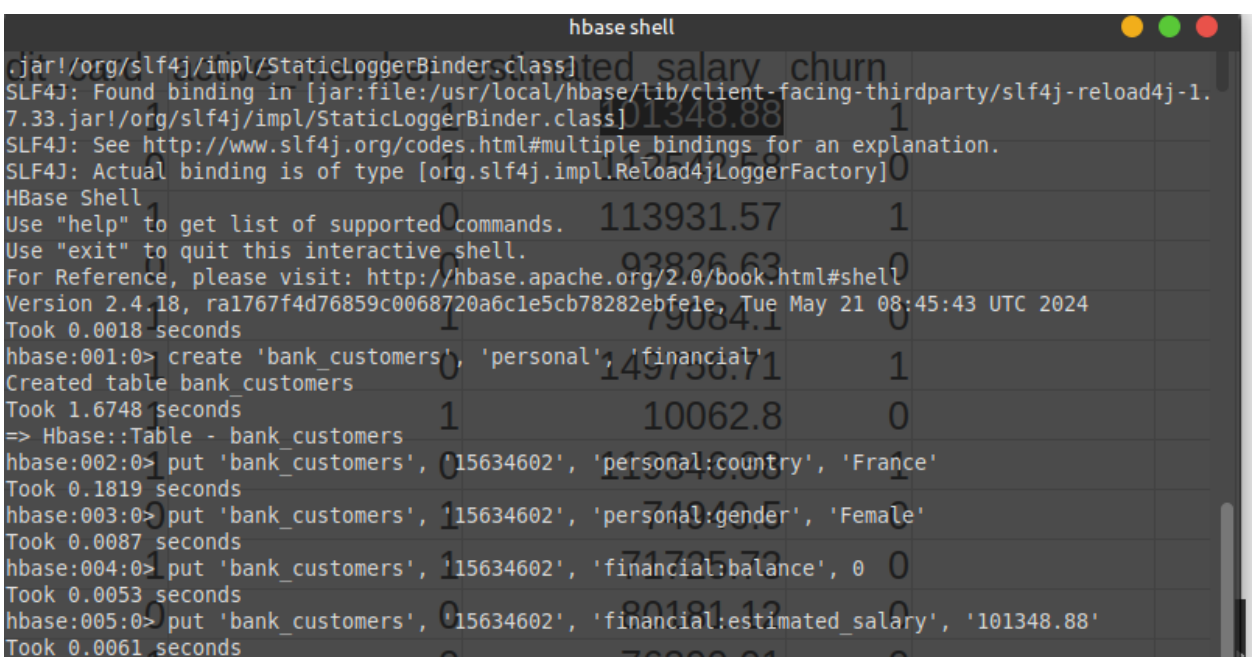
drop 'bank_customers'

Output:

1. HDFS directory file listing



2. HBase table creation with column families



```
hbase shell
hbase:023:0> describe 'bank_customers'
Table bank_customers is ENABLED
bank_customers
COLUMN FAMILIES DESCRIPTION
{NAME => 'financial', BLOOMFILTER => 'ROW', IN_MEMORY => 'false', VERSIONS => '1', KEEP_DELETED_C
ELLS => 'FALSE', DATA_BLOCK_ENCODING => 'NONE', COMPRESSION => 'NONE', TTL => 'FOREVER', MIN_VERS
IONS => '0', BLOCKCACHE => 'true', BLOCKSIZE => '65536', REPLICATION_SCOPE => '0'}
```

```
hbase shell
jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/local/hbase/lib/client-facing-thirdparty/slf4j-reload4j-1.
7.33.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Reload4jLoggerFactory]
HBase Shell
Use "help" to get list of supported commands.
Use "exit" to quit this interactive shell.
For Reference, please visit: http://hbase.apache.org/2.0/book.html#shell
Version 2.4.18, ra1767f4d76859c0068720a6c1e5cb78282ebfe1e, Tue May 21 08:45:43 UTC 2024
Took 0.0018 seconds
hbase:001:0> create 'bank_customers', 'personal', 'financial'
Created table bank_customers
Took 1.6748 seconds
=> Hbase::Table - bank_customers
hbase:002:0> put 'bank_customers', '15634602', 'personal:country', 'France'
Took 0.1819 seconds
hbase:003:0> put 'bank_customers', '15634602', 'personal:gender', 'Female'
Took 0.0087 seconds
hbase:004:0> put 'bank_customers', '15634602', 'financial:balance', 0
Took 0.0053 seconds
hbase:005:0> put 'bank_customers', '15634602', 'financial:estimated_salary', '101348.88'
Took 0.0061 seconds
hbase:006:0>
```

3.2 Retrieval

```
hbase shell
hbase:006:0> get 'bank_customers', '15634602'
COLUMN                          CELL
 financial:balance              timestamp=2026-02-03T23:09:30.629, value=0
 financial:estimated_sala       timestamp=2026-02-03T23:11:27.408, value=101348.88
 ry
 personal:country               timestamp=2026-02-03T23:06:56.743, value=France
 personal:gender                timestamp=2026-02-03T23:08:29.673, value=Female
1 row(s)
Took 0.0682 seconds
hbase:007:0>
```

```
hbase:016:0> scan 'bank_customers'
ROW                    COLUMN+CELL
 15634602              column=financial:balance, timestamp=2026-02-03T23:09:30.629, value=0
 15634602              column=financial:estimated_salary, timestamp=2026-02-03T23:11:27.408, v
                       alue=101348.88
 15634602              column=personal:country, timestamp=2026-02-03T23:06:56.743, value=Franc
                       e
 15634602              column=personal:gender, timestamp=2026-02-03T23:08:29.673, value=Female
 15647311              column=financial:balance, timestamp=2026-02-03T23:15:53.915, value=8380
                       7.86
 15647311              column=financial:estimated_salary, timestamp=2026-02-03T23:16:12.032, v
                       alue=112542.58
 15647311              column=personal:country, timestamp=2026-02-03T23:14:50.833, value=Spain
 15647311              column=personal:gender, timestamp=2026-02-03T23:15:02.754, value=Female
2 row(s)
Took 0.0074 seconds
hbase:017:0>
```

3.3 Updation

```
hbase:017:0> put 'bank_customers', '15634602', 'financial:balance', '78000'
Took 0.0120 seconds
hbase:018:0> put 'bank_customers', '15634602','financial:balance'

ERROR: wrong number of arguments (given 3, expected 4..6)

For usage try 'help "put"'

Took 0.0104 seconds
hbase:019:0> get 'bank_customers', '15634602','financial:balance'
COLUMN                 CELL
 financial:balance     timestamp=2026-02-03T23:18:02.682, value=78000
1 row(s)
Took 0.0105 seconds
```

3.4 Deletion

hbase shell

```
hbase:020:0> delete 'bank_customers', '15634602', 'financial:estimated_salary'
Took 0.0110 seconds
hbase:021:0> get 'bank_customers', '15634602','financial:estimated_salary'
COLUMN                    CELL
0 row(s)
Took 0.0078 seconds
hbase:022:0> get 'bank_customers', '15634602'
COLUMN                    CELL
 financial:balance        timestamp=2026-02-03T23:18:02.682, value=78000
 personal:country         timestamp=2026-02-03T23:06:56.743, value=France
 personal:gender          timestamp=2026-02-03T23:08:29.673, value=Female
1 row(s)
Took 0.0047 seconds
```